An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

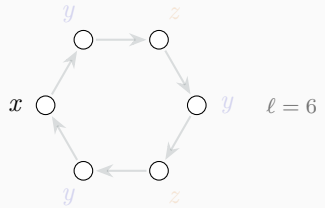
Yuya Uezato Yoshiki Nakamura

CyberAgent, Inc. Chiba University

STOC 2027 submission (author video for the PC)

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



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$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

☛ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

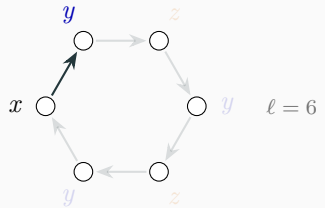
スライド概要：FO^k は変数を使い回せる。モデル検査は $O(|\varphi| n^k)$。

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- 辺を 1 本ずつ出し、式の対応する部分と一緒に読む
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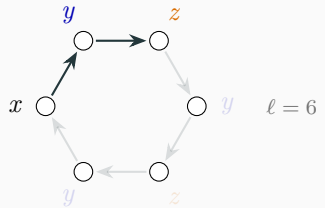
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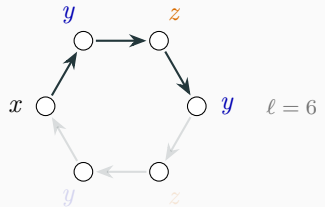
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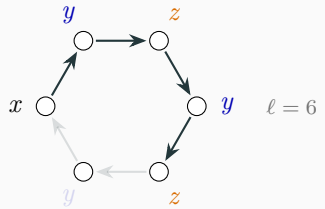
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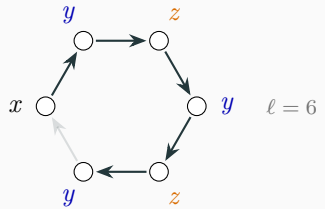
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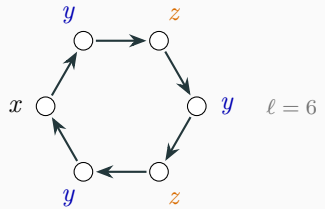
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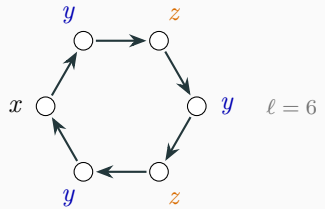
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We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under HyperClique Hypothesis	[BS21, ICALP] [NU27, PODS]
FO^3	$2^{O(\lvert\varphi\rvert)} \cdot n^\omega$ no $2^{o(\lvert\varphi\rvert)} \cdot n^{\omega-\varepsilon}$ under 3-OV Hypothesis	[NU27, PODS] [Wil14, CSL-LICS, for PNF] [NU27, PODS]
CoR [Tar41, JSL] Tarski's calculus of relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z.\ - \wedge -$, we use DNF translation.
$\rightsquigarrow$ $2^{O(\lvert\varphi\rvert)}$ compositions are used [Nak22].

スライド概要：$k \geq 4$ では n^k が最適、FO^3 だけ n^ω に下がる。代償は翻訳の $2^{O(|\varphi|)}$。

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Are $2^{O(|\varphi|)}$ compositions unavoidable?

DNF translation is used, for example:

$$\exists z.\ \big(P_1(x,z) \vee Q_1(z,y)\big) \wedge \cdots \wedge \big(P_k(x,z) \vee Q_k(z,y)\big)$$

$$\rightsquigarrow_{\text{(DNF)}}\ \exists z.\ \bigvee_{A \subseteq [k]} \Big(\bigwedge_{i \in A} P_i(x,z) \wedge \bigwedge_{i \notin A} Q_i(z,y) \Big)$$

$$\rightsquigarrow_{\text{(to CoR)}}\ \Big(\bigcup_{A \subseteq [k]} \Big(\bigcap_{i \in A} P_i \Big) ; \Big(\bigcap_{i \notin A} Q_i \Big) \Big)(x,y)$$

(In this translation, we use 2^k compositions)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS]**)**

Can we prove that there is no $2^{o(|\varphi|)}$-size translation from FO^3 to CoR?

Partial answer: Yes for $\Sigma_1\mathrm{CoR}$ (with transitive closure) [Nak20, RAMiCS; Nak22, JLAMP]

The argument of [Nak22, JLAMP] is based on the fact that $\Sigma_1\mathrm{CoR}$ is equivalent to **existential** FO^3, which is closed under **extensions**.

スライド概要：この指数個の合成は必要か？ 未解決だった（$\Sigma_1\mathrm{CoR}$ のみ既知）。

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The answer is **YES!**

Thm. (This paper)

There is no $2^{o(|\varphi|)}$-size translation from FO^3 to CoR.

Thm. (Also in this paper)

For each k, there is

1. *an FO^3 sentence of size $O(k)$ that is FO^2-definable, but every equivalent FO^2 sentence has $\geq 2^k$ quantifiers;*
   before: FO^{k+1} vs. FO^k known for $k \geq 3$ [VS24, LICS]; FO^3 vs. FO^2 only on linear orders [GS05, LMCS]
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👉 Our proof technique can apply to all the above!

スライド概要：答えは YES（主定理）。同じ手法で FO^2、行列言語、閉包付き CoR にも指数下界。

メモ：

- FO^2：3 変数は 2 変数より指数的に簡潔（線形順序では多項式）
- 行列言語：Brijder–Gyssens–Van den Bussche の問いに答える
- 変わるのは「1 演算が気づける変更の数」だけ（量化子 1、集約 3β、閉包 1）
- 論文には他に：CoR 等式、逆なし回路（ちょうど 2^k）、乱択回路、ランダム入力

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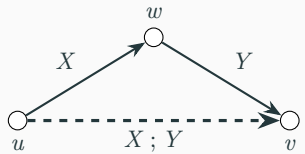
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The calculus of relations (CoR)

$\mathbf{0},\ \mathbf{1},\ \mathbf{id},\qquad \cup,\ \cap,\ \breve{\ },\ \breve{\ },\qquad$ and **composition** ;



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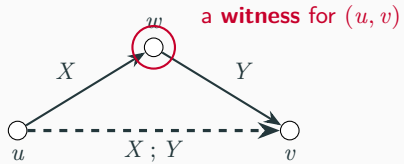
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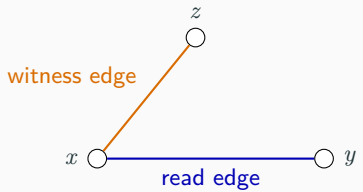
1. Main results

2. Technical overview

3. Conclusion

The formula G_k

$$G_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k} \left[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z)) \right]$$



at every coordinate i: one edge meets P_i, the other meets Q_i $\Rightarrow$ the read edge **accepts** the witness edge

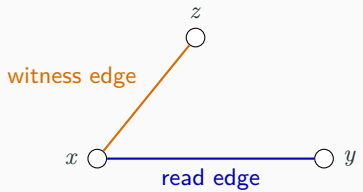
スライド概要：G_k の読み方：read edge が witness edge を受理する。

メモ：

- 各座標で一方が P_i、他方が Q_i
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	$i=1$	$i=2$	$i=3$	$i=4$
read edge	P_1	Q_2	P_3	Q_4
witness edge	Q_1	P_2	Q_3	P_4

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The DNF: 2^k compositions

$$\underbrace{H_\emptyset \cap (H_{[k]}\,;\mathbf{1}) \quad \cup \quad H_{\{1\}} \cap (H_{\{1\}^c}\,;\mathbf{1}) \quad \cup \quad H_{\{1,2\}} \cap (H_{\{1,2\}^c}\,;\mathbf{1}) \quad \cup \quad \cdots}_{G_k \equiv \bigcup_{A\subseteq[k]} \left[H_A \cap (H_{A^c}\,;\mathbf{1})\right] \;:\; 2^k \text{ compositions}}$$

$H_A \coloneqq \bigcap_{i\in A} P_i \cap \bigcap_{i\notin A} Q_i$; a read edge in H_A accepts every witness edge in H_{A^c}

Is the DNF optimal?

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Main results

Thm. (Exponential lower bounds for full CoR)

Every relational circuit $C \equiv_{\mathrm{fin}} G_k$ has $\geq 2^{k-1}$ composition gates.

- shared subterms, complement, converse: all allowed and free
- also: term size $\Theta(k2^k)$; the blowup begins at domain size $2^{\Theta(k)}$

スライド概要：主定理：合成は 2^{k-1} 個以上必要。

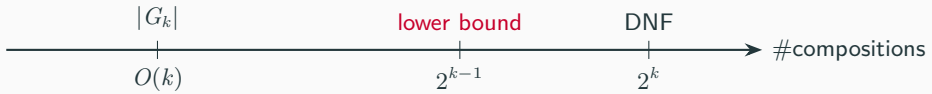
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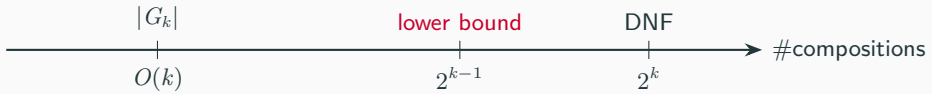
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Why complement makes it hard

Exponential lower bounds were known only for $\Sigma_1\mathrm{CoR}$ (complement only on atoms) [Nak22, JLAMP].

	$\Sigma_1\mathrm{CoR}$	full CoR
closure	under extensions	none ($\overline{R\,;1}$ fails once an R-edge is added)
fooling set	composition separates ≤ 1 instance	two compositions separate all

👉 Our invariant is an **equality**, not an inclusion: it survives complement.

スライド概要：既存手法は補で壊れる。包含ではなく等式を使う。

メモ：

- $\Sigma_1\mathrm{CoR}$ は拡張で保存 → fooling set
- 補があると閉包性が崩れる

Outline

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Our technique: exact predictions on one reference structure

$\mathcal{M}_0$

only **compositions** can notice a modification

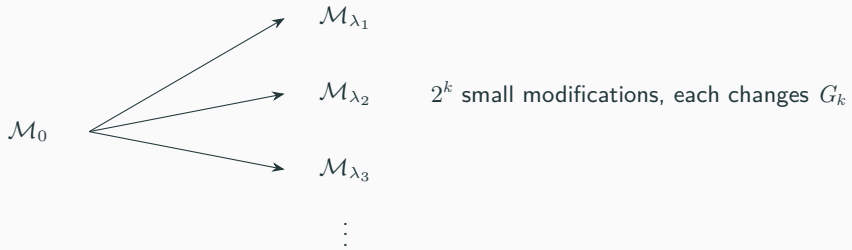
each composition notices at most **two** $\Rightarrow$ $2^k \leq 2 \cdot g_c(C)$

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- 以降のページはこの 3 行の中身

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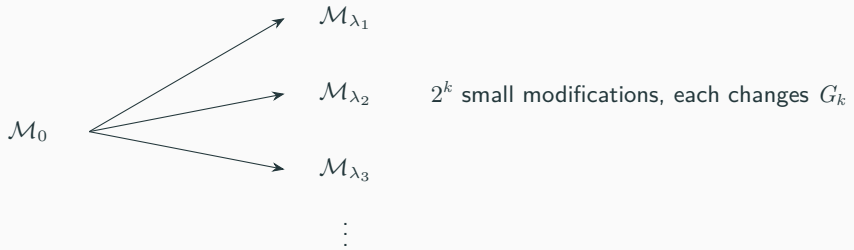
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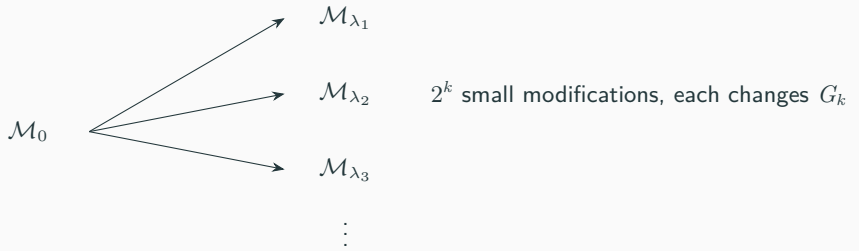
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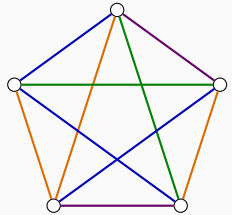
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The reference structure



an edge-colored complete graph

colors $\mu \subseteq [k]$; on an edge of color μ:

$$P_i \iff i \in \mu, \qquad Q_i \iff i \notin \mu$$

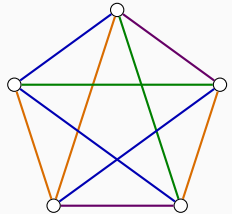
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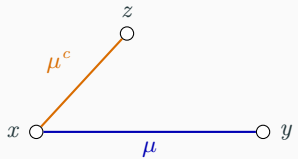
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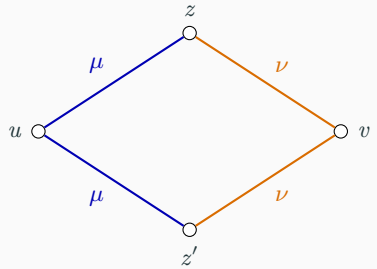
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for all $u \neq v$ and all colors μ, ν: at least two witnesses

exists for $n \geq n_0(2^k)$ (random coloring) [JMT95, Algebra Univers.]

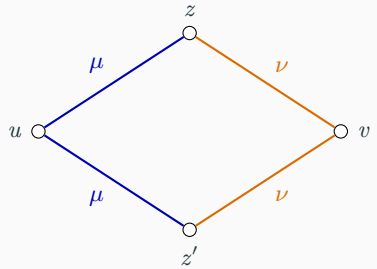
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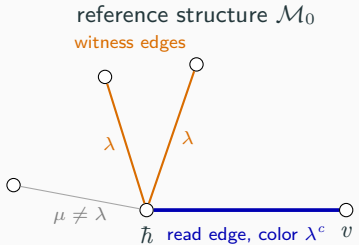
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$$G_k(\hbar, v) = 1$$

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the fold of λ recolors **those edges at the hub**, and nothing else

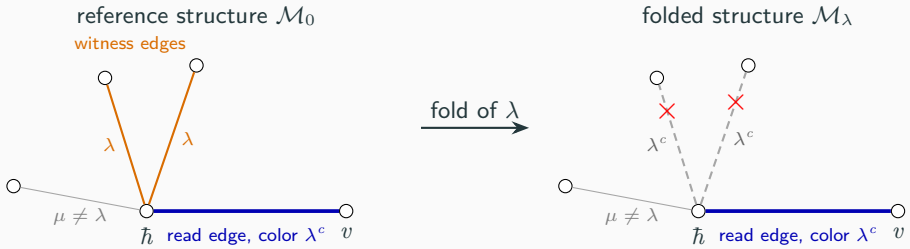
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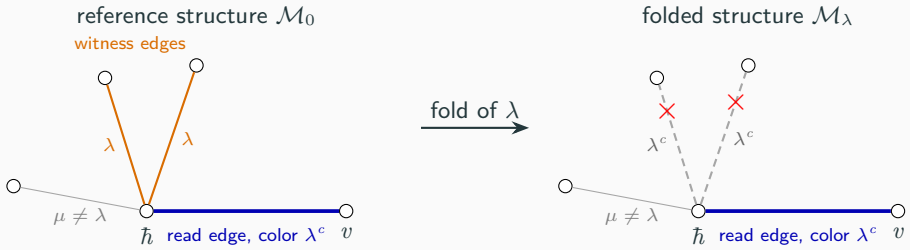
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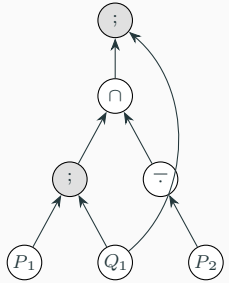
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Predictions and noticing



a relational circuit

gate by gate: is $[\![g]\!]_{\mathcal{M}_\lambda}$ what $[\![g]\!]_{\mathcal{M}_0}$ **predicts**? $(\star)$

Δ_g: the folds for which g violates $(\star)$

g **notices** λ: a new violation at g

Boolean operations and converse notice nothing

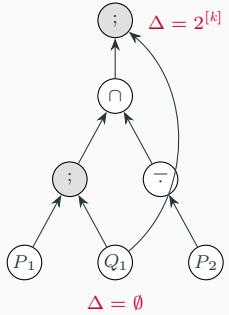
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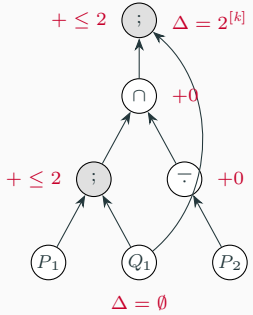
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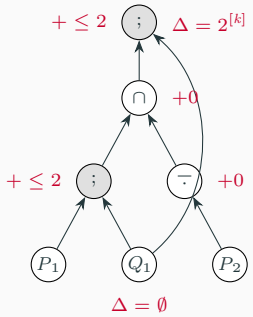
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every relation computed on $\mathcal{M}_0$ has a **color table** (however much complement is used)

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Lem. (Each composition notices at most two folds)

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on an arbitrary structure: a single $R\,;\,S$ reacts to t separate changes

on the reference structure: at most two

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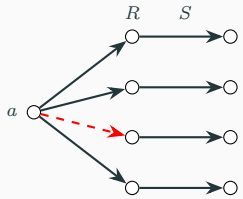
メモ：

- 一般の構造なら t 個に反応しうる

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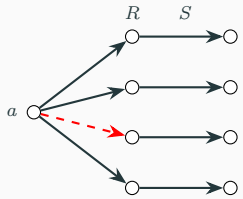
メモ：

- 一般の構造なら t 個に反応しうる

Each composition notices at most two folds

Lem. (Each composition notices at most two folds)

For color tables X, Y: $(X\,;\,Y)^{(\lambda)} = X^{(\lambda)}\,;\,Y^{(\lambda)}$ for all λ except at most two.



on an arbitrary structure: a single $R\,;\,S$ reacts to t separate changes

on the reference structure: at most two

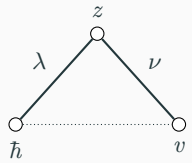
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At the hub, and away from the hub

(a) $\mathcal{M}_0$, at $(\hbar, v)$



witness z: $(X\,;\,Y)(\hbar, v) = 1$

$X \coloneqq H_\lambda$, $Y \coloneqq H_\nu$ $(\nu \neq \lambda)$; noticed: the prediction $(X\,;\,Y)^{(\lambda)} = 1$ differs from $X^{(\lambda)}\,;\,Y^{(\lambda)} = 0$

differ only if $S_X = \{\lambda\}$, $S_Y = \{\lambda\}$, or $S_X \cap S_Y = \{\lambda\}$ $\Rightarrow$ at most two λ

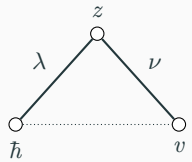
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- hub の外では hub を避ける witness z' が残る
- 条件は 3 つ、どれも λ を決める $\rightarrow$ 高々 2

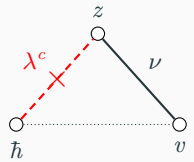
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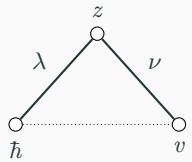
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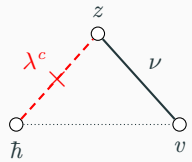
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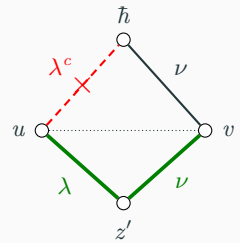
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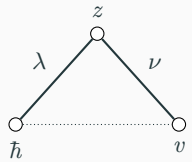
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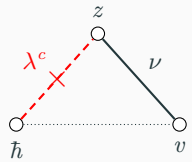
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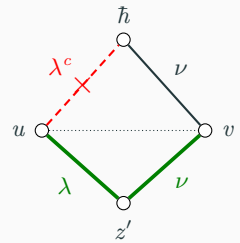
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Outline

1. Main results

2. Technical overview

3. Conclusion

Summary

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- some FO^3 formulas of size $O(k)$ need $\geq 2^{k-1}$ compositions in every equivalent CoR term, even with complement, converse, and shared subterms;
- the same technique: FO^3 vs. FO^2, matrix query languages, CoR with closure.

Technique 2^k folds of one reference structure; each composition notices ≤ 2 of them.

? Open questions

- a superpolynomial gap over a **fixed** finite signature (e.g., one binary relation)?
- the domain size from which the blowup begins?
- the exact number of compositions for G_k: between 2^{k-1} and 2^k?

Thank you for your attention!

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メモ：

- 未解決：固定シグネチャ、開始の領域サイズ、正確な合成数

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